Africa’s Tropical Forests

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Photo 1: Evidence of logging along the Mpivié river.

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Introduction

In 2018 I went on an expedition to the Loango coastal forest region of southwestern Gabon as a voluntary field agent under the auspices of Stuart Nixon, Chester Zoo's Regional Field Programme Manager. I would like to thank Stuart and Charlie Childs for taking care of me on that expedition and everyone else I met who inspired and supported me while I wrote this review. The contents are split into four parts to understand how forest conservation is tied to wildlife, climate, and human existence in Central Africa.

- Part 1 - The history of forests and gorillas in equatorial Africa to the modern day.
- Part 2 - Tropical forest interactions with a changing climate, hydrology, and humans.
- Part 3 - Governments, peoples, and communities utilisation and valuation of forests.
- Part 4 - Complexities, methods (e.g. ecosystem services), and contexts for conservation.

Photo 2: Adult antlion in a grassland patch of secondary rainforest mosaic near Fernan Vaz lagoon.

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Part 1 — Forests and Gorillas

Central Africa

The Central Africa region includes the countries of Cameroon, Gabon, Equatorial Guinea, the Democratic Republic of the Congo (DRC), the Republic of Congo (Congo), the Central African Republic (CAR), and the Angolan exclave of Cabinda (Abernethy et al., 2016). The region covers around 2.28 million km² (Hoare, 2007) and includes parts of central and southern Cameroon, the CAR, Republic of Congo, Gabon, Equatorial Guinea, and the DRC (Kimbu, 2011). Dense rainforest presides across this equatorial belt region but not equally (White, 2001a; Abernethy et al., 2016) as we shall see. The Congolian rainforests extend from the eastern Albertine Rift Valley to the western coast and include areas in DRC, southern CAR, Congo, and Cabinda (Abernethy et al., 2016).

As for forests, the Congo Basin contains 1.8 million km² of forest that reaches across DRC, most of Congo, the southeastern parts of Cameroon, the southern CAR, as well as Gabon and Equatorial Guinea (Nogherotto et al., 2013). The basin is home to the second largest (Kleinschroth et al., 2019) and most intact tropical rainforest after the Amazon (Nkem et al., 2013) and represents approximately 20% of the world's remaining tropical forest (Mayaux et al., 2004). The most extensive stretch of rainforest encompasses the Congo Basin, the adjoining lower Guinean region, and smaller block in West Africa (White, 2001a).

To the west of the Congo River's drainage, the contiguous Lower Guinean forests extend east and south from the Cross River through southern Cameroon, Gabon, and Equatorial Guinea (Abernethy et al., 2016). These forests are collectively referred to as the Guineo-Congolian rainforests but are biologically distinct from the West African rainforests that stretch from southern Senegal to the Dahomey Gap (White, 1985). Additionally, further distinct swamp forests grow along the south Nigerian coasts and Niger River Delta that span the coast from the Dahomey Gap to the Cross River (Olson et al., 2001).

Paleo-history

The world transitioned from the Mesozoic to the Cenozoic Era 66 million years before-present (Ma YBP) and we are now in the seventh epoch (Cohen et al., 2023). Starting from the Paleocene (66.00 Ma YBP), through to the Eocene (56 Ma YBP), Oligocene (33.9 Ma YBP), Miocene (23.03 Ma YBP), Pliocene (5.333 Ma YBP), Pleistocene (2.58 Ma YBP), up to our present Holocene (11,700 YBP) epoch (Cohen et al., 2023). During the Eocene epoch about 50 Ma YBP, CO_2 rose to over 1000 ppm and it was the last time forests reached their greatest extent (Willis & McElwain, 2014), then by the Oligocene CO_2 had fallen and grasses began replacing forests (Bond & Midgley, 2012). During the same time, C4 type grasses evolved that, unlike their ancestral C3 pathway, grow well in conditions of high heat, drought, salinity (Sage, 2004), sunlight, and fire frequency (Ratnam et al., 2011).

The Last Glacial Maximum (LGM) is the last time ice sheets reached their greatest extent and lasted until around 10,000 YBP (Maley, 1996). During the LGM equatorial Africa experienced a significant drop in temperature (Maley, 2001), approximately 4 °C lower than current levels (Bonnefille et al., 1990; Tierney et al., 2008) and CO_2 was much lower than during interglacial periods (Bond & Midgley, 2012). Pollen studies from Lake Bosumtwi in Ghana and Lake Barombi Mbo in West Cameroon suggest the LGM occurred between 20,000 and 15,000 YBP (e.g. Maley 1989, 1991) (Maley, 1996). But upon different evidence it is posited that the LGM may have occurred anywhere between 26,500 to 19,000 YBP (Clark et al., 2009a) or 18,000 to 12,000 YBP (Maley, 2001). In addition to lower temperatures during the LGM, monsoon circulation was weakened and moisture transport from the Atlantic Ocean was diminished (Gasse, 2000; Schefuß et al., 2005; Tierney et al., 2008). These effects collectively led to a decrease in rainfall (Bonnefille and Chalie, 2000) and a subsequent increase in aridity (Hamilton, 1982), that in turn caused forests to reduce in extent (Maley, 1996) [*see Part 2 - Forests and Climate*].

When there is a reduction in forest extent grasses move in to colonise, and with a smaller leaf area index than trees, evapotranspiration decreases and causes surface albedo to increase (Nogherotto et al., 2013) (i.e. increased radiation reflectance). Evergreen rainforests seemingly expand during wet intervals and shrink during periods of drought (Ngomanda et al., 2007). Moreover, today's Central African forests originate from islands of 'refugia' that managed to persist throughout the LGM (Maley, 1996, 2001; Willis et al., 2013; Städele et al., 2022). Therefore, the oldest forests are located within former refugia dating to over 18,000 YBP (Abernethy et al., 2016) that include the massive swamps in central DRC and the mountain ranges of the western equatorial forests (Abernethy et al., 2016). Conversely, to complete the picture, the youngest forests are only some decades old and can be found hundreds of kilometres north and south of the refugia, transitioning to and engulfing savannahs (White, 2001b).

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Gorillas

Looking more closely at gorillas, Western gorillas are genetically and morphologically distinct from gorillas found 1000 kilometres to the east with two named species in each group; as supported by a comprehensive literature review spanning 85 years (Tocheri et al., 2016). It is estimated that western and eastern gorillas diverged only 1.2 to 3.0 Ma YBP (Thalmann et al., 2007; Becquet and Przeworski, 2007) with gene flow continuing up to around 200,000 to 80,000 YBP (Thalmann et al., 2007), or possibly as recently as 20,000 YBP (Becquet & Przeworski, 2007; Mailund et al., 2012; McManus et al., 2015; Scally et al., 2012; Thalmann et al., 2007; Xue et al., 2015), indicating a gradual separation (Städele et al., 2022). Interestingly, genetic studies of western gorillas exhibit significantly larger past effective population sizes, greater genetic diversity, and a more pronounced population genetic substructure compared to their eastern counterparts (Thalmann et al., 2007, 2011; Scally et al., 2012, 2013; Prado-Martinez et al., 2013; Roy et al., 2014a; Xue et al., 2015; Fünfstück and Vigilant, 2015).

Western

In western Africa, **Western lowland gorillas** (*Gorilla gorilla gorilla*) span western central Sub-Saharan Africa and have the largest geographical distribution of all gorilla subspecies, with an estimated population size of 362,000 individuals (Strindberg et al., 2018). **Cross River gorillas** (*G. g. diehli*) are another western gorilla subspecies that inhabit a highly fragmented and confined area straddling the border between Cameroon and Nigeria (Städele et al., 2022), with the smallest estimated population size of the gorilla species ranging between 200 to 300 (Dunn et al. 2014).

The limited and fragmented Cross River gorilla habitats likely contribute to reduced genetic diversity and greater differentiation (propagation of certain traits within the population) due to limited gene flow and mating opportunities (Städele et al., 2022). Cross River gorillas diverged from Western lowland gorillas between 17,800 (Thalmann et al., 2011) and 68,000 YBP (McManus et al., 2015; Prado☐Martinez et al., 2013; Thalmann et al., 2011) during a long cold period that lasted from 75,000 to 10,000 YBP during which forest regression was significant (Maley, 1996). However, gene flow between the two groups did not stop till approximately 420 YBP and was followed by a genetic bottleneck in Cross River gorillas just 320 YBP (Thalmann et al., 2011).

Eastern

In eastern Africa, **Mountain gorillas** (*G. beringei beringei*) are only found in two small, discrete populations, totalling around 1,000 individuals in the Virunga Massif spanning Rwanda, Uganda, and the DRC and including Bwindi Impenetrable National Park in Uganda (Granjon et al., 2020). **Grauer or Eastern lowland gorillas** (*G. b. graueri*) are located in the eastern DRC and they have the second-largest spatial distribution among the gorilla subspecies with an estimated population of approximately 6,800 individuals (Plumptre et al., 2021). Grauers are thought to have split from mountain gorillas between 20,000 (Jensen☐Seaman & Kidd, 2001; Roy et al., 2014a) and 10,000 YBP (Roy et al., 2014a).

Mountain gorillas, along with some Grauer gorilla populations inhabit high-altitude areas characterised by afromontane forests (Mehlman, 2008). Lowland variety Grauers consume a larger amount of fruit and climb more often, with more extensive and seasonally variable ranging patterns aligning more closely with western lowland gorillas (Casimir, 1975; Goodall and Groves, 1977; Yamagiwa et al., 1992; Yamagiwa and Mwanza, 1994; Uchida, 1996; Yamagiwa et al., 1996; Doran and McNeilage, 1998, 2001; Robbins and McNeilage, 2003). Yet, despite having a larger distribution than mountain gorillas, Grauer's genetic diversity and differentiation still suffers from potential geographic and environmental barriers (Städele et al., 2022).

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Photo 3: Long-legged katydid in logged primary rainforest along the Mpivié river.

Habitat

Gorillas diverged from the evolutionary line leading to modern humans, chimpanzees, and bonobos only between 12.0-8.8 Ma YBP (Scally et al., 2012). The middle Miocene occurred 15.98 Ma YBP (Cohen et al., 2023) and during this time East Africa's dense forest cover gave way to a mosaic landscape of forest and savannah (e.g. Van Zinderen Bakker & Mercer 1986; Hamilton 1991; Harris 1993) (Maley, 1996). The upper Miocene started 11.63 Ma YBP (Cohen et al., 2023) as indicated by the introduction of Asian and European flora and fauna to Africa 10.5 Ma YBP, whereby the African climatic zonation began to resemble present-day patterns (Maley, 1996). During the upper Miocene about 8.0 Ma YBP, C4 grasses became very prominent (Cerling et al., 1997) possibly facilitated by low CO_2 levels (Ehleringer et al., 1997; Osborne, 2008; Edwards et al., 2010) such as which occurred in East Africa during the last glaciation (Wooller et al., 2000; Damsté et al., 2011).

Fast-forward to the LGM and the landscape of the highlands along the Albertine Rift is transformed with an arid glacial climate and lower atmospheric CO_2 concentrations that depress montane forest belts by over 1,500 metres (Hamilton, 1982). That led to the advancement of glaciers in the Rwenzori mountains to elevations as low as 2,000 metres above sea level, 2,700 metres lower than today (Livingstone, 1967; Kelly et al., 2014). On East African mountains such as the Rwenzori, these LGM conditions allowed for alpine grasslands to become the dominant vegetation type until about 15,000 YBP (Livingstone, 1967) when the early deglacial period began (Maley, 1996). These cold and arid conditions in east Africa during the LGM made gorilla habitation largely unsuitable with active populations being confined to refugia; as demonstrated in an extensive literature review (Tocheri et al., 2016).

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The African Humid Period (AHP) then lasted from 15,000-5,000 years YBP (de Menocal et al., 2000) or possibly 11,000-4,000 YBP (Willis et al., 2013). The AHP brought with it increased temperatures and precipitation, punctuated by the near-glacial Younger Dryas period between 12,800-11,700 YBP (Zhang et al., 2014). The conclusion of the AHP 5000 YBP (e.g. de Menocal et al., 2000; Tierney et al., 2008) marked a period of rainforest decline and the expansion of grasslands and savannahs in the lowlands (Beuning et al., 1997). In East Africa moderate drying is noted at 6,200-5,500 YBP that abruptly shifts to arid conditions at 4,900-4,500 YBP (Tierney et al., 2010).

Increased precipitation during the AHP (e.g. Tierney et al., 2011; Kageyama et al., 2013) allowed for forests to remerge that likely provided the necessary habitat for Grauers to split from mountain gorillas either just before or just after the Younger Dryas period (Tocheri et al., 2016). The AHP reached its maximum at 8,500 YBP and during this time tropical forest species are noted as reaching up into the Sahara/ Sahelian by growing along watercourses (Watrin et al., 2009). The proto-Grauer may have dispersed westward from the Virunga forests alongside an expanding forest habitat into what is now known as the grauer gorilla range, located in the highlands northwest of Lake Kivu (Tocheri et al., 2016).

The late Holocene then starts 4,200 YBP to the present day (Cohen et al., 2023) during which rainforests across western equatorial Africa are then characterised by successive millennial-scale changes (Maley & Brenac, 1998; Vincens et al., 1998; Giresse et al., 1994, 2005; Delègue et al., 2001; Wirrmann et al., 2001; Nguetsop et al., 2004) driven by natural climatic variability (Vincens et al., 1999; Elenga et al., 2004). Increasing aridity from 4,000 to 1,300 YBP caused a further reduction in African lowland rainforests (Ngomanda et al., 2007) that largely resulted in the modern vegetational landscape (Vincens et al., 1999).

Tropical rainforests extended their range 1,200 to 800 YBP that may be attributed to increased moisture during the "Medieval Warm Period" (MWP) ~1,100 to 800 YBP (Ngomanda et al., 2007) that unevenly affected parts of the globe (Bradley, 2000; Broecker, 2002). For example, from 3,200 YBP onwards West Africa is noted as undergoing a decrease in wetness and an increase in temperature (Schefuß et al., 2005). Between 1,300 to 600 YBP equatorial west Africa experienced low water levels that may be due to decreased rainfall (e.g. Nguetsop et al., 2004; Wirrmann et al., 2001; Giresse et al., 2005) (Ngomanda et al., 2007) during the MWP.

Introduction of Humans

In parts of Central and central West Africa 2,500 YBP an arid event is registered that may have caused forest retreat (Vincens et al., 1999; Maley, 2002) that has also been attributed to human induced land-use change (e.g. Neumann et al., 2012) (Bayon et al., 2012). From 2,700 YBP East African tree-like species have declined and grasses have increased (Vincens, 1993). In Uganda of East Africa 2,000 YBP, a decline in montane tree taxa and rise in grass pollen indicates human-induced deforestation (Livingstone, 1967; Taylor, 1990) and the human clearance of the Virunga forests are documented to have begun around 1,000 YBP (McGlynn et al., 2013).

Over the past 1000 years there has been minor variation in the Guineo-Congolian rainforest cover suggesting that global-scale climatic variability has had little effect (Ngomanda et al., 2007). In fact, various pollen records indicate that vegetation fluctuations in equatorial western Africa during this time are resultant from human activities (Elenga et al., 1994; Reynaud-Farrera et al., 1996; Vincens et al., 1998, 1999). However, the data is too patchy to say with certainty at the decadal or centennial resolution and more archaeological investigations are required (Ngomanda et al., 2007). In my opinion, this is inconsequential to the contemporary evidence explored later.

Gorilla Extinction

Gorillas live in groups where reproduction is dominated by one (a silverback) or a few males (Schaller, 1963; Fossey, 1974; Harcourt, 1979; Robbins, 1995; Watts, 1996; Bradley et al., 2005; Breuer et al., 2010; Masi et al., 2021; Vigilant et al., 2015). For example about half of mountain gorilla social groups have multiple resident adult males but the dominant silverback may sire up to 72% of the offspring (Bradley et al., 2005; Vigilant et al., 2015). High male reproductive skew like that has implications for genetic diversity (range of traits within the population) and differentiation (propagation of certain traits within the population) (Tocheri et al., 2016).

Isolation reduces genetic diversity and increases differentiation (Bradley et al., 2005; Breuer et al., 2010), whereas long-distance male dispersal increases genetic diversity and reduces differentiation (Douadi et al., 2007; Roy et al., 2014b; Masiet al., 2021). Research on gorilla morphology, particularly with grauer gorillas, points toward low levels of genetic diversity among eastern gorilla populations (Ruvolo et al., 1994; Garner and Ryder, 1996; Ruvolo, 1997; Saltonstall et al., 1998; Yu et al., 2004; Prado-Martinez et al., 2013). Low genetic diversity adversely impacts adult survivorship (Xue et al., 2015; Fünfstück & Vigilant, 2015), thereby heightening the risk of extinction for gorillas (Tocheri et al., 2016).

Today, Grauer gorilla habitat is being reduced and fragmented by forest clearance, compounded by bushmeat hunting and the Ebola virus outbreak which have drastically reduced present□day population sizes (Walsh et al., 2003; Plumptre et al., 2016; Strindberg et al., 2018). Ebola has also caused a reduction in the western gorilla population size (Walsh et al., 2003; Leroy et al., 2004; Bermejo et al., 2006; Rizkalla et al., 2007). Currently, western lowland gorillas exhibit the greatest genetic diversity across all measures, but a reduction in genetic diversity can affect individual fitness and survivability (Städele et al., 2022). Of mountain and Grauer's gorillas, 34% and 39% of their genomes respectively, are homozygous, which indicates excessive inbreeding (Xue et al., 2015; van der Valk et al., 2019).

Threat of Humans

Deforestation of gorilla habitat and bushmeat trafficking reduce gorilla populations and exacerbate the risks of extinction for gorillas (Harcourt, 1996; Walsh et al., 2003; Yamagiwa, 2003; McNeilage et al., 2006; Robbins et al., 2011). Rural and urban communities in tropical Africa depend significantly on wild meat for protein, fuelling a growing commercial bushmeat trade (Milner-Gullans & Bennnett, 2003). Bushmeat is a primary source of protein for many households, especially forest dependent people who are traditionally not engaged with animal husbandry (Nasi et al., 2011). Grauers, western lowland, and Cross River gorillas are already classified as Critically Endangered, and mountain gorillas as Endangered (Maisels et al., 2018; Plumptre et al., 2019).

Commercial hunting in Central Africa has significantly impacted wildlife populations, such as with the loss of 62% of Central Africa's elephants from 2002 to 2011 (Maisels et al., 2013). With products like ivory becoming scarcer, hunters shift their focus to other species to remain profitable (Fa & Brown, 2009). In the Congo basin, 60% of 57 studied mammal species are being exploited unsustainably (Fa et al., 2002). In West Africa, persistent overhunting has nearly caused the bushmeat trade to collapse, with some primate species facing likely extinction (Oates et al., 2000). Hunters in forest communities generally do not hunt beyond a 10 km radius from a village (Van Vliet, 2010) but as increasing trade opportunities emerge they are being driven to extend further and exert an increasing pressure into the forest (Kümpel et al., 2010). These complexities will be explored further in Part 3.

Part 2 — Forests and Climate

Climate change is arguably the most persistent threat to global stability in the coming century (Adger et al., 2003). Human influence has unequivocally warmed the planet with global surface temperature increasing since 1970 faster than any other 50-year period in at least 2000 years (Allan et al., 2023). The IPCCs Fifth Assessment Report has identified Representative Concentration Pathways (RCPs) to evaluate projected emission scenarios (Lyon et al., 2022). RCP 2.6 is the most low-impact scenario (Stocker et al., 2013) predicting low-emissions into the future (Lyon et al., 2022) and RCP 8.5 is a high-emission trajectory (Van Vuuren et al., 2011). RCP 6.0 falls between these two extremes with global temperatures predicted to increase from 2.2 °C to 3.6 °C between 2100 and 2200 (Lyon et al., 2022).

In tropical rainforests temperatures are expected to increase by 2-5 °C over this century (Hulme et al., 2001; Giorgi, 2002; Cramer et al., 2004). Global warming last reached 5 °C between 11,700-11,500 YBP (Kobashi et al., 2008) with Greenland ice cores even suggesting a 20-60 year period of 10 °C of warming during that time (Steffensen et al., 2008). Already in the last decade [2011-2020] temperatures have exceeded the most recent multi-century warm period of around 6,500 YBP (Allan et al., 2023). Africa is predicted to warm at a faster rate than the global average (James & Washington, 2013), so even under RCP 2.6 overall warming across Central Africa may become over 3 °C by the end of the century (United Nations, 2015).

Empirically observed climatic trends suggest an increased seasonality and heightened water stress across the northern and western Lower Guinean forests that may alter both forest function and extent (Abernethy et al., 2016). For the central, southern, and eastern forests of the Congo Basin, the situation is less clear, but eastern DRC is expected to experience a wetter climate (Zelazowski et al., 2011). A strong drying trend has been observed in the northern Congo, whereas the southern Congo has seen less of a reduction in rainfall (Zhou et al., 2014). Additionally, West Africa has been subjected to two decades of persistent drought and seen a shift towards drier conditions (Nicholson, 1994; Fauset et al., 2012) despite increases in monsoon precipitation since 1980 (Allan et al., 2023).

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Photo 4: Thread-waisted wasp (Genus Chlorion) on Fernan Vaz lagoon's banks.

The Congolese forest is drier than the rainforests of Amazonia and may be more tolerant of short-term rainfall reduction than wetter tropical forests (Asefi-Najafabady & Saatchi, 2013). Also, Certain Central African trees possess a higher resilience compared to their neotropical (New World) counterparts that is attributed to their history of enduring post-Pleistocene epoch drought periods (Willis et al., 2013) (e.g. Vincens et al., 1999). For example, rainforest soils such as those of the Lake Nguène catchment in Gabon can be hydromorphic, which may have allowed them to survive through decadal wet-dry episodes (Ngomanda et al., 2007). A relatively high proportion of large, old evergreen trees with high wood density add to forest resilience, but may also be most vulnerable to an ecological tipping point resulting from physiological stress (Lindenmayer et al., 2012).

Drying may be associated with forest retreat (Malhi et al., 2008), particularly at 4°C (James et al., 2013) and especially since drier conditions heighten the risk of forest-fires (Nepstad et al., 2004; Randerson et al., 2005; Poulter et al., 2010). The rainforests of central Africa have been experiencing long-term drying conditions (Malhi & Wright, 2004; Asefi-Najafabady & Saatchi, 2013) and northern African tropics have been experiencing rapid warming since at least 1970 (Malhi & Wright, 2004). There has been a persistent browning in the Congolese forests that may be a reflection of a gradual adjustment to long-term drying, rather than the sporadic droughts seen in the Amazon (Zhou et al., 2014). A strong drying trend has also been observed in the central (Malhi & Wright, 2004; Asefi-Najafabady & Saatchi, 2013), western (Malhi & Wright, 2004; James et al., 2013), and northern African tropics, despite an overall increase of precipitation in East Africa and globally (James et al., 2013).

Limiting Factors

Soil water is the most important factor determining the distribution of forest, savannah, and grassland but is depleted through transpiration (Tinley, 1982). It is over timescales of thousands to tens of thousands of years that soil generally improves in suitability for supporting plant growth (Van Breemen, 1993). The soil parent material influences vegetation and soil chemistry more so than in-situ tree species do (Augusto et al., 2003). That said, the distribution of trees in a forest can have a large bearing on the amelioration, or degradation of its soil (Zinke, 1961; Binkley & Giardina, 1998; Augusto et al., 2003; Frelich et al., 2015). In deforested land both soil moisture and evapotranspiration decline, after which the soil is less able to extract and use moisture (Nogherotto et al., 2013). Highly flammable C4 grasses are the first to colonise cleared forests and recurrent grass-fuelled fires can strongly suppress tree regrowth (Bond et al., 2003). Grasses typically have a higher surface albedo than forests from retaining dead reflective leaves, but transpire less water due to their smaller roots, leaf area, and canopy roughness (Bond & Midgley, 2012).

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Increasing photosynthetic active radiation (energy) can enhance radiation-limited canopy photosynthesis (Nemani et al., 2003; Huete et al., 2006) and evapotranspiration, while energy remains the limiting factor (Woodward, 2008; Zhou et al., 2014). But, when soil moisture becomes the limiting factor, increasing energy may enhance water stress for plant growth (Zhou et al., 2014). Water stress results in less turgid (wilting) leaves, to their shedding altogether, that in the short term is recoverable once rainfall returns (Zhou et al., 2014). Thinning and/ or retreating forests then reduce evapotranspiration and increase run-off, resulting in decreases to precipitation rates (James et al., 2013). This may be accelerated as higher temperatures increase evapotranspiration (Woodward, 2008) and decrease productivity (Delire et al., 2008). Long-term ground observations and drought manipulative experiments are needed to understand the impacts of climate change on forests (Nepstad et al., 2007) but research of this kind is not available for Congolese forests (Zhou et al., 2014).

Water deficit over an extended period of time may alter forest species composition and structure from a gradual loss of photosynthetic capacity and water content (Lewis, 2006; Enquist & Enquist, 2011; Fauset, 2012) that may have knock-on effects for biodiversity and carbon storage (Lewis, 2006; Valentini et al., 2014). Long-term drought can reach a critically low threshold of water availability whereby closed-canopy forests transition to more open, lower-biomass forests (Lewis, 2006; Brncic et al., 2007; Enquist & Enquist, 2011; Fauset et al., 2012). A strong drying trend may also cause the photosynthetic capacity of trees and their moisture content to be reduced, which could favour the spread of drought-tolerant species (Lewis, 2006; Delire et al., 2008; Enquist & Enquist, 2011). As seen in Ghanaian forest species shifting from wetter evergreen trees towards drier deciduous species (Fauset et al., 2012). Once open land is established, higher rainfall promotes the growth of wooded savannah (Sankaran et al., 2005) instead of forests, which differ in their species composition, structure, and function (Ratnam et al., 2011).

Atmospheric Effects

Forests have a low surface albedo and so contribute to planetary warming through increased heat absorption on land, but they also cool the climate through evapotranspiration as part of the hydrologic cycle (Bonan, 2008). The level of radiation required for plant growth and transpiration is determined by atmospheric conditions (clouds and aerosols) and latitude (Zhou et al., 2014). In this instance, surface warming associated with the low albedo of forests is offset by strong evaporative cooling from transpiration, but can become inhibited via drought (Bonan, 2008). Inversely, increased land evapotranspiration due to human-induced climate change has contributed to increases in agricultural and ecological droughts (Allan et al., 2023).

The combined effects tropical deforestation has on surface albedo, evapotranspiration, and cloud cover may accelerate global-scale warming in equatorial tropical forests (Bala et al., 2007). Observed water deficits across Central Africa are driven by factors such as lower rainfall, higher post-storm runoff (Beyene et al., 2012), and increased evapotranspiration due to warmer temperatures (Woodward, 2008; Abernethy et al., 2016). Notwithstanding, reduced evapotranspiration in a drier climate may lead to the loss of tropical forests (Baldocchi et al., 2000). Making biosphere evapotranspiration, surface roughness, and albedo all key controls in the general circulation of the atmosphere which could affect the regional climate (Hayden, 1998).

Regional climates in Central Africa have been identified as particularly sensitive to the exchange of water and energy between the surface, the atmosphere (Koster et al., 2004) and biogeophysical processes (Bonan, 2008). In particular, extreme events like cyclones, fire weather, heatwaves, heavy precipitation, and droughts are expected to increase in frequency and intensity in direct relation to increasing global warming (Allan et al., 2023). Whereas the impact of changes in precipitation over short and long timescales on tropical rainforests remains poorly understood (Malhi & Wright, 2004; Brando et al., 2010; Atkinson et al., 2011; Xu et al., 2011; Samanta et al., 2012; Asefi-Najafabady & Saatchi, 2013; Saatchi et al., 2013; Morton et al., 2014).

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Carbon Dioxide

The central African evergreen tropical forests act as a carbon sink (Saatchi et al., 2011a), with the Congo Basin alone estimated to contain between 25 and 30 billion tons of carbon (Hoare, 2007). Over the last six decades, land and ocean have consistently taken-up about 56% per year of human induced CO_2 emissions but this capacity will decrease under higher CO_2 emissions (Allan et al., 2023). Plants can be influenced by CO_2 by altering water use efficiency, photosynthetic rates, light and nutrient efficiency, and the relative performance of C3 and C4 photosynthesis (Ratnam et al., 2011). Decreased CO_2 during the LGM drove the retreat of forests and expansion of flammable grasslands (Wooler et al., 2000). But, in 2019, global CO_2 concentrations were higher than at any time in at least 2 million years and CH_4 and N_2O were the highest in at least 800,000 years (Allan et al., 2023). Increasing CO_2 can result in a CO_2 fertilisation process whereby alongside warming, photosynthesis and plant productivity are enhanced (Norby et al., 2005; Woodward, 2008; Willis & Bhagwat, 2009; Willis & MacDonald, 2011).

Increasing CO_2 also often results in reduced transpiration/ leaf area that slows the depletion of soil water (Polley et al., 1997) and favours the recruitment of tree seedlings in place of savannah grasses (Bond & Midgley, 2012). Unlike tropical forests, the evaporative cooling generated by temperate and boreal forests have "little to none" and "counterproductive" mitigative effects on global warming respectively (Bala et al., 2007). Such that, when temperature increases in boreal forests terrestrial net primary production (vegetation growth) is enhanced, whereas it decreases for tropical forests due to greater evaporative demand that dries soil (Fung et al., 2005). Despite this, in recent decades tropical forests have shown an increase in productivity, tree growth, recruitment, mortality rates, and forest biomass (Lewis et al., 2009) probably due to increases in CO_2 (Lloyd & Farquhar, 2008). Subsequently, global warming may exacerbate under a positive feedback loop between decreased evaporative cooling and the amount of released CO_2 (Betts et al., 2004).

Larger forest trees are much more resistant to fire than their smaller counterparts (Bond & Midgley, 2012) and as larger trees establish themselves in place of grasses, there will be large impacts on fluxes of water, nutrients, carbon, and even weathering (Pagani et al., 2009). However, forest trees and savannah trees are considered as alternative ecosystem states between pyrophobic (fire repellant) and pyrophytic (fire resistant) respectively (Warman & Moles, 2009; Lehmann et al., 2011; Staver et al., 2011). Wildfires can play a natural part of the carbon cycle whereby CO_2 uptake during the growing season alleviates the emissions from fires (Valentini et al., 2014). Increasing CO_2 has indeed resulted in young savannah trees relative to 100 YBP, possessing faster growth, stronger regrowth, massive root systems, and larger starch reserves (Bond & Midgley, 2012). Since savannahs cover a large spatial extent, even a small increase in woody biomass would result in a substantially increased carbon sink (Archer et al., 2001; Asner et al., 2003; Williams et al., 2005).

Photo 5: Hairy Golden Orb-weaving Spider (Trichonephila fenestrata) in secondary rainforest

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Climate model projections predict a drier Atlantic coast that may result in further decreases of rainforest extent (Delire et al., 2008). Large-scale conversion of tropical forest to pasture also creates a warmer, drier climate (Bonan, 2008) and between 1980 and 2000, nearly 60% of new agricultural land in Africa was derived from intact forests (Gibbs et al., 2010). Compared with pastureland, tropical forests maintain high rates of evapotranspiration, lesser surface air temperature, and greater precipitation (Bonan, 2008). Water stress affects the spatiotemporal probability of rapid land use changes, particularly regarding the suitability of land for agricultural purposes (Abernethy et al., 2016). Moreover, deforestation is coupled with the local and global climate, and as drier regions are more likely to be deforested (Malhi & Wright, 2004) global warming is accelerated (Cox et al., 2000). Therefore, in Central Africa climate change has the potential to accelerate forest loss through land conversion to agriculture (Laurance et al., 2014; Bele et al., 2015).

Forest Loss

A reduction in the absolute range of a species will likely increase the risk of local extinction (Thomas et al., 2004; Thuiller et al., 2005) and tropical rainforests are vanishing more rapidly than any other biome (Laurance, 1999; Achard et al., 2002). In Central Africa the main drivers of deforestation may be attributed to clearance for smallholder agriculture (Damania & Wheeler, 2015), firewood and charcoal extraction (Mayaux et al., 2013), timber logging (often illegally) (Hoare, 2015), and mining activities (Edwards et al., 2014a). In addition to the forest habitat, native wildlife is also under threat from industrial logging, slash-and-burn farming, overhunting, and increased infrastructure development (Laurance et al., 2006).

A significant cause of deforestation in central Africa is the combination of rapid population growth and extensive road building (Wilkie & Laporte, 2001; Naughton-Treves & Weber, 2001). The Congo Basin is critically impacted by the effects of road construction (Kleinschroth et al., 2019) with the World Bank predicting that over the next 15 years it will be the driving factor for deforestation in the basin (Megevand, 2013). The extraction of timber and precious minerals start near urban centres and major transport routes and pave the way for lower value extraction such as charcoal and smallholder agriculture (Ahrends et al., 2010). As seen within 10-15 km of villages and roads, whereby wildlife populations, such as duikers (forest antelope), bushpigs, elephants, and diurnal primates, experience significant declines (Barnes et al,. 1991; Lahm et al., 1998; Fa et al., 2000; Wilkie et al,. 2000; Lahm, 2001).

Over time, concentric haloes of deforestation form around cities and transport routes that drive the demand for yet more resources (Abernethy et al., 2016). Beyond a critical density of 8 people per square kilometre canopy loss accelerates and deforestation haloes around villages become permanent (Mayaux et al., 2013). Few species can survive in the mosaic of farmland and degraded scrub left behind from a large-scale forest invasion (Laurance, 2001a). These haloes may even merge in regions with high rural population densities such as parts of the DRC, Equatorial Guinea, and southern Cameroon (Abernethy et al., 2016). Road construction also serves as a gateway for oftentimes illegal mining, land colonisation, poaching, and exacerbated bushmeat hunting (Laurance et al., 2009; Barber et al., 2014; Kleinschroth & Healey, 2017).

Logging

It is estimated that only 1 person per square kilometre may be supported while sustaining tropical forest biomass (Robinson & Bennett, 2000). In terms of rural populations, all countries in the Central Africa region have surpassed this figure, except for Gabon (Abernethy et al., 2016). The deforestation rate in Central Africa is projected to increase by between 0.3% and 0.5% per year by 2020-2030 (Nogherotto et al., 2013). West Africa has only about 12% of its original rainforest left, Eastern Africa has just 8%, and Central Africa retains about 59% (Naughton-Treves & Weber, 2001). Logging may result in major deforestation (Kleinschroth et al., 2019) and frequently opens up forested areas to be encroached upon by shifting agriculture (Ernst et al., 2013; Molinario et al., 2017), which is the primary cause of forest loss across Africa (Curtis et al., 2018).

Without severe hunting pressure, wildlife may persist in logged forests albeit with reduced abundance (Johns, 1997). Nation states in the Congo Basin grant concessions (long-term leases) to logging companies (Karsenty & Ferron, 2017). For example the DRC has allocated extensive timber leases to Zimbabwe, Germany, Malaysia, and China (Laurance et al., 2006) with the most substantial lease spanning 1.5 times the size of Great Britain, 34 Ma hectares (Laurance, 2001b). Within well-managed concessions outside of DRC, the impact of logging roads on forest loss is mitigated by a fourfold higher rate of road abandonment than in areas outside concessions (Kleinschroth et al., 2019). While concessions do maintain much higher carbon stocks and wildlife habitat than cleared areas can (Goetz et al., 2015) most logging operations in the tropics are poorly managed as reduced-impact logging can be more expensive than traditional methods (Putz et al., 2000).

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Unsustainably logged forests are prone to catastrophic wildfires (Cochrane et al., 1999; Nepstad et al., 1999), especially during droughts (Nepstad et al., 2004; Randerson et al., 2005; Poulter et al., 2010). Whereas selective logging can have positive impacts on tropical ecosystems (Laurance & Laurance, 1996; Johns, 1997; Malcolm & Ray, 2000; Fimbel et al., 2001;) and within concessions can provide economic value so long as they remain forested (Putz et al., 2012; Edwards et al., 2014b). However, results of a study across an array of African tropical forests, have found that over the course of 30 years selective logging significantly reduces carbon levels in forests (Valentini et al., 2014). Furthermore, the maze of tracks left by selective logging serves to enhance accessibility into the forested areas by hunters, miners, and farmers that subsequently destroy or severely degrade forests (Wilkie et al., 1992, 2001; Laurance, 2001a). Nevertheless, reduced impact logging should be promoted to reduce damage to the forest and maintain biodiversity (Sonwa et al., 2011).

Part 3 — People of the Forest

Forests are crucial to the well-being of a large number of poor people (Arnold & Ruiz-Pérez, 1998; Byron & Arnold 1999; Sunderlin et al., 2005; Tieguhong & Ndoye, 2006; Powell et al., 2011; Wan et al., 2011) yet they are rarely considered in national planning and policy (Bele et al., 2015) and neither are indigenous communities (Salick & Ross, 2018; Wiben et al., 2019). Across the Congo basin's 113 Ma population, many of those in rural settings still maintain a partly hunter-gatherer lifestyle, along with small-scale shift-and-burn agriculture for subsistence (Abernethy et al., 2016). The DRC contains anywhere between 600,000 to 2 million indigenous peoples, Burundi and Chad are home to hundreds of thousands, and Cameroon, Rwanda, and the CAR range in the tens of thousands at least (Wiben et al., 2019).

"Pygmies" are a collection of indigenous groups within the Congo Basin with a unique culture at risk of extinction (Kimbu, 2011). 'Pygmy' (Pygmée) is a stereotypical term (Moïse, 2011), but is used here to describe those who are directly dependent on the forest (Nkem et al., 2013). However, it should be noted that reductive categorisations such as "hunter-gatherer"/ "farmer" and "pygmy"/ "villager" transform the narrative into contrasting subsistence strategies and polarised relationships of power (Woodburn, 1997; Rupp, 2003). The Mbuti, Baka, and Batwa/ Twa peoples of Central Africa consider the term "Pygmies" derogatory and discriminatory (Wiben et al., 2019). Even reference to 'nomadic lifestyles' can be considered problematic when it comes to determining indigenous peoples' rights to property and land (Vinding, 2019). Therefore, long standing Pygmy relations with their 'exploiters' is important to consider when developing their platform for autonomy (Moïse, 2011).

Pygmies must use forests other than those bordering their settlements, which brings them into contact with other communities that influence their livelihood strategies such as with bushmeat sales (Nkem et al., 2013). They easily become victims of discrimination (Woodburn, 1997; Hewlett, 2000; Lewis, 2000; Jackson & Payne, 2003) and the powerful (Rupp, 2003, 2011) as they are lacking in social status (Lewis, 2000; Klieman, 2003) material wealth, development, and basic human and citizen rights (Hewlett, 2000). The Baka speaking people live in the tropical forests of southern Cameroon, northern Congo, and Gabon (figure 1) but are exploited by Bantu farmers, villagers, and government officials (Hewlett, 2000). Notwithstanding, the Batwa peoples in the Tanganyika province of DRC face issues of violence and displacement and rebel groups in CAR are subjugating Pygmies to enact unsustainable hunting (Wiben et al., 2019).

Forest Resources

Forests provide refuges for biodiversity, food, medicinal, and forest products, regulation of the hydrological cycle, protection of soil resources, recreational uses, spiritual needs, and aesthetic values (Bonan, 2008). Millions of people are dependent on the forest for their livelihoods (Sunderlin et al., 2005) medicinal plants, and other resources (World Bank, 2004). Many rural communities in the Congo Basin also rely on shifting agriculture and forest resources for their livelihoods (Pelletier et al., 2018). Therefore forests play a crucial role in reducing the vulnerability of natural and social systems (Bele et al., 2015).

Many developing countries' national development plans and poverty reduction strategies must depend on tropical-forests as natural resource pools (Smith & Scherr, 2003; Nkem et al. 2010; 2013). Forests are used for in-situ forest hunting (Nasi et al., 2011) and gathering (Bahuchet, 1994; Wan et al., 2011), such as the forest communities of southern Cameroon whom depend on non-timber forest products to survive (Ndoye et al., 1998; van den Berg & Biesbrouck 2000; Sassen & Jum, 2007; Nkem et al. 2010). However, the provision of tropical-forest ecosystem goods and services (bushmeat, crops, timber, herbal medicine, honey, etc...) are rapidly declining and are predicted to be severely altered due to climate change impacts (Bele et al., 2015) such as global warming (Delire et al., 2008).

It is highly dispersed communities, predominantly indigenous ones, that are most vulnerable to the impacts of climate change (Nkem et al., 2010; Bele et al., 2011). For example, aforementioned CO_2 fertilisation effects can contribute to woody thickening in savannahs and cause land users to work much harder to maintain ecosystems such as grasslands (Bond & Midgley, 2012). Notwithstanding, some forms of small-scale agriculture result in an over-extraction of biomass that leads to land degradation and lower crop yields, which then further exacerbates the agricultural pressure on the land (Bennett et al., 2009). Mining, mechanised-agriculture, logging, and over-exploitation of forests for fuel-wood also threaten the livelihoods of forest-dwellers (Pearce, 2012). If forest communities' vulnerability to climate variability is addressed, the negative impacts of climate change may be reduced (Ribot et al., 1996).

Adaptation and Mitigation

Mitigation loosely defined reduces the accumulation of greenhouse gases (e.g. by enhancing carbon sinks), while adaptation reduces the vulnerability of societies and ecosystems to climate change (Ravindranath, 2007; Locatelli et al., 2008). Climate change impacts can either be mitigated by reducing greenhouse gases and enhancing carbon sinks (e.g. forests) (Klein et al., 2005; Allan et al., 2023) or adapted for by adjusting natural or human systems to ensure a sustainable continuation of peoples livelihoods (Ravindranath, 2007). While mitigation has global benefits and may well be implemented on the same local or regional scale, adaptation typically works on the scale of an impacted system, which is regional at best, but mostly local (Klein et al., 2005).

The United Nations Framework Convention on Climate Change has identified adaptation and mitigation (A/M) as potential policy responses to climate change (Klein et al., 2005; Ravindranath, 2007; Somorin et al., 2012). Adaptation is paramount as the poorest regions are anticipated to be hardest hit (Adger et al., 2003; Tol et al., 2004; Thomas & Twyman, 2005; Bele et al., 2011) but society cannot adapt indefinitely (King, 2004). Yet for many developing countries, adaptation is a priority due to a high vulnerability to climate impacts (Ayers and Huq, 2009; Somorin, 2010). Therefore, in 2015 the IPCCs Sixth Assessment Report used Shared Socio-economic Pathways (SSPs) to model different climate policy scenarios while integrating the challenges of A/M (Allan et al., 2023).

People’s perspective on adaptation shapes the discourse around it and vice-versa, meaning adaptation policy is sensitive to the way in which it is perceived (Somorin et al., 2012). Climate change will affect temperature and rainfall patterns, which will expose forest-dependent groups to physical, mental, and health risks (Costello et al., 2009). Many forest communities have deep-grained social structures (van den Berg & Biesbrouck, 2000) and are relatively isolated, which makes adaptation difficult or impracticable without causing disruption (Nkem et al., 2013). Improperly implemented adaptation actions e.g. draining wetlands or moving peoples, could damage natural ecosystems and invoke maladaptation in the long-term (Campbell, 2009).

Developed countries contain the vast majority of the world’s scientific and technological capacity, so it is in their interest to help developing countries leapfrog into non–carbon emissions technologies (King, 2004). State, market, and civil institutions play an important role in determining the response to conditions and risks associated with the environment and policy (Adger, 2000; Adger and Vincent, 2005; Smit and Wandel, 2006). Mechanisms such as the Global Environmental Facility and Kyoto Protocol’s Adaptation Fund support adaptation strategies (Ravindranath, 2007). However, the Clean Development Mechanism, the Kyoto Protocol, and other global mechanisms mostly focus on mitigation (Ravindranath, 2007).

Photo 6: Netwing Beetle (Genus Lycus) in logged primary rainforest along the Mpivié river.

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Mitigation Mechanisms

Economic studies indicate that reducing tropical deforestation is cost-effective for climate change mitigation (Enkvist et al., 2007; Strassburg et al., 2012). And, in 2005 Gabon and New Guinea spearheaded the REDD+ framework introduced by the UN Framework Convention on Climate Change, which was then internationally adopted at the Paris COP21 in 2015 (Abernethy et al., 2016). The framework aims to establish a global system for rewarding carbon storage and forest preservation (Abernethy et al., 2016) whereby an additional 1% of protected forests in Central Africa equates to over USD~$500M (Bele et al., 2015). Such an approach has the potential to reduce damaging deforestation (O'Sullivan, 2009) and thereby reduce the vulnerability of forest-dependent communities to climate change (Brown et al., 2010).

Forest preservation and conservation are mitigation measures (Kremen et al., 2000; Thompson et al., 2009) that generate carbon credits (Somorin et al., 2012). The Congo Basin governments have shown a preference towards policy agenda such as REDD+ since it provides a source of revenue (Somorin et al., 2012). Investing in small-scale technologies that enhance soil moisture can reverse land degradation, curb erosion, boost carbon sequestration, and increase crop yields simultaneously (Enfors & Gordon 2007; Enfors et al. 2008). Payments for environmental services could then be used to supplement agricultural approaches and technologies that reduce emissions (Campbell, 2009).

Forests are able to sequester large amounts of carbon annually (Bonan, 2008) and REDD+ acts both to encourage carbon sequestration while simultaneously reducing our reliance upon natural forests (Somorin et al., 2012). However, while forests may be globally valued for climate regulation (Saatchi et al., 2011a), keeping forests can reduce the ability of rural households to escape poverty (Campbell, 2009). Forests may be locally valued for fuelwood but timber harvesting by one individual can lead to habitat decline, impacting another person's access to bushmeat (Fisher et al., 2009). Poorly implemented, the REDD+ framework could deprive people of the main source for sustaining their livelihoods (Locatelli et al., 2008) or endanger previously undisturbed forest types (Nkem et at., 2013).

Logging is expected to cause severe and potentially irreversible damage within the next decade (Bele et al. 2015) but the private forestry sector is generally against adaptation and consider it instead to be a responsibility of the government (Brown et al., 2010; Somorin et al., 2012). Moreover, the economic returns of industrial logging creates more immediate wealth than conservation initiatives (Kremen et al., 2000). The $5 per ton rate for avoided carbon emissions set by REDD+ is significantly lower than competitive pricing against other land uses (Abernethy et al., 2016). REDD may work as a financial incentive for shifting cultivators and extensive cattle ranchers in the humid tropics, but not for commercial agricultural crops, logging, or even smallholder intensive agroforestry (Campbell, 2009).

Future land transformation in Africa is unknown, but a number of socioeconomic drivers may either increase or decrease pressure on the expansion of land-use activities (Thuiller et al., 2006). Mitigation efforts tend to operate within the context of international obligations, whereas adaptation is usually more local in nature, concerning economies and livelihoods (Martens et al., 2009). REDD+ can be said to target local communities (Somorin et al., 2012) as adaptation opportunities exist in forest conservation, afforestation, and reforestation (Ravindranath, 2007). But these solutions are ultimately framed within the context of a mitigation agenda (Campbell, 2009).

Adaptation Strategy

National adaptation policies and programmes focused on forest dependent-communities should consider livelihood strategies and natural resource management (Sonwa et al., 2011; 2012). In Pygmy societies human development could choose to focus on autonomy (e.g. in education, hunting, gathering, agriculture, and wage labour), entrepreneurialism (e.g. as a diviner/ healer, or master hunter), and creativity/ play (e.g. storytelling, comedic performances, graphic arts) (Moïse, 2011). The effects of climate change on the natural resource base must also be evaluated (Nkem et al., 2013). Therefore, better downscaled climate models are needed for seasonal forecasting and longer term planning under new climate regimes (Campbell, 2009).

The UNFCCC's 'New Collective Quantified Goal on Climate Finance' funding (UNFCCC, 2015) has been criticised for not meeting targets or the needs of those most affected by climate change and lacking understanding of climate finance (Pauw, 2025). No mitigation activity should reduce adaptation and so the global effort should attempt to focus on A/M synergies (Ravindranath, 2007). Moreover, international conventions have been known to deal with interrelated issues such as desertification, biodiversity loss, and climate change separately (Cowie et al., 2007). Our understanding may be improved by an interdisciplinary framework combining climate change science, climate impacts on ecosystems, and climate change mitigation policy (Bonan, 2008).

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Institutions and networks allow for fair power distribution and so can enhance adaptive capacity to climate change (Tompkins and Adger, 2004; Walker et al., 2006; Pahl-Wostl, 2009). For instance, Cameroon's adaptive capacity would be enhanced by institutional linkages and coordination across the government, private sector and civil society (Brown et al., 2010). An integrated approach to A/M may combine adaptation and mitigation where synergies exist e.g. using REDD+ income to finance adaptation, but then treat A/M separately where trade-offs are inevitable (Somorin et al., 2012). To achieve a more integrated approach to A/M new forms of engagement between scientists, policy-makers, and wider stakeholder communities are required (Martens et al., 2009).

Governance & Tourism (Gabon & Cameroon)

Gabon is roughly the size of Italy at 268,000 km² and is situated on the equator just south of Cameroon (Laurance et al., 2006). The country contains dense rainforest, swamps, mangroves, steppes, savannahs (Lahm, 2001) and about 40% of the world's gorillas (Walsh, 2006). Ecotourism aims to improve protected area management and provide economic benefits to local residents but struggles to compete with the competitive pricing of timber companies for access to old-growth trees (Wilkie & Carpenter, 1999). Gabon designated thirteen national parks with the aim of creating a sustainable ecotourism model (Laurance et al., 2006) and unlike Cameroon, has an efficient tourism development, promotion, and marketing framework in place (Kimbu, 2011).

Photo 7: Variegated grasshopper (Zonocerus variegatus) within smallholder agriculture.

In the 1990s the French oil company Elf made substantial payments to political leaders in countries including Gabon and Cameroon (Henley, 2003). Crude oil exports contribute to over 40% of Gabon's gross domestic product (GDP) but are declining, which may result in increased timber exploitation (Laurance et al., 2006). Logging practices in Gabon frequently result in significant environmental damage due to inadequate regulation (Laurance et al., 2006), leading to extensive changes in forest structure, microclimatic shifts, soil compaction, and erosion (Fimbel et al., 2001; Johns, 1997). In Gabon alone less than 3% of the 221 registered logging companies have even submitted the required management plans by law (Laurance et al., 2006).

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Cameroon covers 475,440 km² (Molua, 2007), situated on the west coast of Africa, it is the leading economy in the Central African region (Harilal et al., 2019) and is often described as an “Africa in miniature” (Mbenda et al., 2014). Cameroon has been ranked as one of the world’s most corrupt according to the Transparency International Corruption Perception Index (Alemagi & Kozak, 2010). Despite this, the country receives the highest number of international visitors in the subregion compared to other Central African countries (Kimbu, 2011). Pygmies represent ~0.4% of Cameroon's 24 Ma population, with the most prominent being the 40,000 Baka living in the east and southern regions but the Bagyeli/ Bakola [Kola] and Bedzang [Medzan] are also present across southern Cameroon (figure 1) (Wiben et al., 2019). Cameroon has over 20 protected areas and many local communities depend on the natural resources within these protected areas for livelihood sustenance and cultural practices (Tchindjang et al., 2005).

Therefore, restricting local access to resources in protected areas may affect forest dependent communities' source of livelihood (Brown et al., 2010) and may undermine support for tourism initiatives (Harilal et al., 2019). In DRC the Kahuzi Biega National Park has been criticised for the expulsion and permanent exclusion of the Batwa indigenous peoples (Wiben et al., 2019). Cameroon validated the REDD+ National Strategy in 2018 following the creation of the ‘Platform REDD+ and Indigenous Peoples of Cameroon’ (Wiben et al., 2019). Then, over the next 2000 years global mean sea level will rise by 2 to 3 metres if warmining is limited to just 1.5 °C, but could reach 22 metres under 5 °C of warming (Allan et al., 2023). Gabon and Cameroon had already surpassed over 50% of their population living in urban areas by 1985 (Abernethy et al., 2016) and even under RCP 2.6, rising sea levels may result in the loss of coastal forests and wetlands, costing billions of dollars by the end of the century (Brown et al., 2011).

Viability of Ecotourism

Central African nations have commonly experienced civil wars, political coups, rebellions, and harassment from police, immigration, and customs officers (Wilkie & Carpenter, 1999). Political instability, recurring wars, and widespread corruption pose challenges for forest conservation (Laurance et al., 2006). For example, the mass displacements and violence in CAR (Wiben et al., 2019) and the DRC (Samset, 2002; Chilunjika, 2024). Additionally, Central African government representatives have been known to be unprofessional and uncourteous (Hewlett, 2000). And, in the period leading up to 2011, the Central Africa subregion trailed behind Eastern and Southern Africa in terms of growth in tourism (Kimbu, 2011).

Gabon and to some extent Cameroon stand out as not having experienced violent internal conflicts over the past fifty years (Abernethy et al., 2016). Since 2004, the Cameroonian government has aimed to reduce its reliance on oil and other extractive industries by developing other sectors such as tourism (Kimbu, 2011). Which has become increasingly difficult since northern Cameroon has been subjected to terrorist attacks from groups such as Boko Haram (Tichaawa, 2017). Fifteen percent of Cameroon's land area is dedicated to nature conservation areas (Riley & Riley, 2005) and maintain relative peace and political stability (Shackley, 2007), which lays a solid foundation for developing a dynamic ecotourism industry (Shackley, 2007; Harilal et al., 2019).

Ecotourism can become a major source of revenue for protected areas if certain criteria are met as historically seen with Gorilla tourism in the DRC, Rwanda, and Uganda (Wilkie & Carpenter, 1999). However, the revenue from ecotourism may be insufficient to cover management costs as seen with the Dzanga-Sangha Special Reserve in the CAR, and the Lope Reserve in Gabon (Wilkie & Carpenter, 1999). For ecotourism to be successful in Cameroon a sustainable tourism industry must be present (Harilal et al., 2019) which requires advancements in infrastructure and collaboration amongst varied stakeholders (Murphy et al., 2000; Khadaroo & Seetanah, 2007; Kimbu, 2011).

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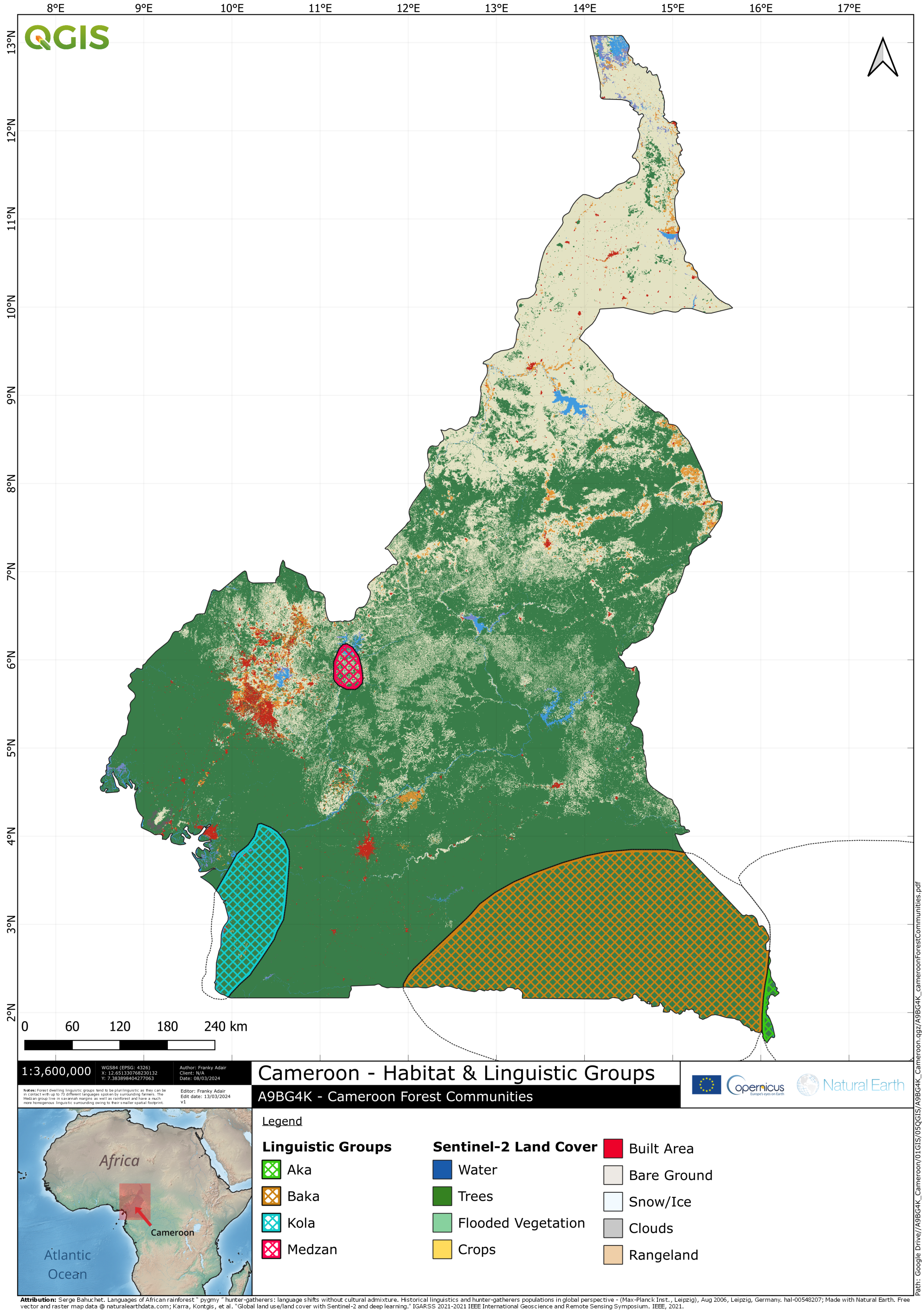

Figure 1: Map of Cameroon depicting Sentinel-2 Land Cover and forest dwelling linguistic groups.

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Part 4 — Ecosystem Conservation

Ecosystem Services (ES) may be loosely defined as the benefits derived from ecosystems (Bennett et al., 2009). That said, humans have engineered many ecosystems for the efficient production of services such as food, timber, and fibre (Foley et al. 2005; Kareiva et al. 2007; Monfreda et al. 2008; Ramankutty et al. 2008). Tropical and subtropical African biomes such as forest and savannah are examples of important ES that carry associated human benefits (Willis et al., 2013). African forests provide regulating services, such as carbon sequestration (Saatchi et al., 2011a) and provisioning services, such as timber, non-timber forest products (Hoare, 2007), and biofuels (Walter & Segerstedt, 2012). In East Africa it is noted that climate change threatens to significantly impact the provision of ES into the future (Conway et al., 2005). ES is therefore an anthropocentric term tied to human existence and so concerns stakeholders across all of society (Fisher et al., 2009).

The Millennium Ecosystem Assessment (MA) is a global study concerning ES that tries to classify services into supporting, regulating, provisioning, and cultural categories (Alcamo et al., 2005). Fisher et al., (2009) states that the MA's definition of ES therefore includes cultural meaning, recreation, and spiritual fulfilment. This can result in errors such as double-counting in overlapping categories like in both the drinking water and spiritual fulfilment derived from a sacred forest (Fisher et al., 2009). Therefore, ES may alternatively be categorised by Organisation (ecosystems), Operation (process/ functioning of ecosystems), and Outcome (benefit provided to humans) (Fisher et al., 2009): For example, the pattern of water stress across Central African forests will play a crucial role in shaping the future of land use in the region (Bell et al., 2015). Forest cover is tied to soil moisture, which is influenced by rainfall and evapotranspiration (Zelazowski et al., 2011). In this case the Organisation is forests, an Operation is rainfall and evapotranspiration, and an Outcome is soil moisture levels.

By simply analysing trade-offs and synergies between ESs at a single point in time or only examining spatial concordance, it risks misunderstanding the underlying mechanisms at play (Bennett et al., 2009). ES can exhibit spatio-temporal dynamism, whereby an example might be water regulation services provided by a mountain forest which will benefit downstream areas over time (Fisher et al., 2009). Moisture availability in the root zone of plants is determined by the sum of rainfall, the level of runoff and evapotranspiration (Reager & Famiglietti, 2013). Notwithstanding, the trees of some forests have deep roots that grant access to groundwater that may delay the effects of water stress (Zhou et al., 2014) but inhibit their expansion (Drake et al., 2011). Therefore, maintaining relationships between ES are crucial for bolstering their resilience and reducing their decline (Bennett et al., 2009) yet the MA found that 15 out of 24 studied ES were declining globally (Fisher et al., 2009).

Focusing only on a limited set of ES such as soil moisture alone, risks regime shifts that could result in the unexpected and sudden loss of other ES (Gordon et al., 2008). As an example, pollination is considered an indirect ES and is crucial for the reproduction of many plants, including crops like almonds (Fisher et al., 2009) and coffee (Ricketts et al., 2004). Integrating forest patches in agricultural landscapes can improve pollination and pest control, which in turn improves crop yield (e.g. Olschewski et al., 2006; Bodin et al. 2006; Priess et al., 2007; Kremen et al. 2007; Ricketts et al. 2008) (Bennett et al., 2009). Small forest patches of moderately-managed landscapes have been demonstrated to conserve tree diversity and forest structure (Hernández-Ruedas et al., 2014). Planting trees also enhances carbon sequestration, but the presence of trees can increase evapotranspiration which may lead to exacerbated water stress (Fahey & Jackson 1997; Engel et al. 2005). Meaning that already dry agricultural plots could actually suffer from reduced moisture levels by planting forest patches nearby.

A major limitation of ES comes in the form of our own limited understanding of their workings (Kremen & Ostfeld, 2005; Carpenter et al. 2009) and of biodiversity itself (Midgley & Thuiller, 2005). Ecosystems exhibit complex lagged relationships between rainfall, vegetation phenology and photosynthetic activity (Lewis, 2006; Enquist & Enquist, 2011; Lewis et al., 2013) and large spatiotemporal variabilities in rainfall (Malhi & Wright, 2004). In Africa forest against savannah distribution doesn't change over a rainfall gradient, which suggests that the two vegetation types are largely determined by local processes (Staver et al., 2011). ES are characterised by feedback loops, time lags, and nested phenomena (Limburg et al., 2002) and differing ecosystems have differing responses from various plant species to drought (Lucht et al., 2006).

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Classifying Spatial Costs

Diminishing environmental quality has a disproportionate impact on those marginalised by the market economy (Dasgupta, 2001; Adger et al., 2003; Nkem et al., 2010). However, the complexity is amplified when differing interests in ES across various spatial scales are considered (Hein et al., 2006). For example, a crucial aspect of ES is in understanding how they contribute to human welfare and equity (e.g. Aylward & Barbier, 1992) (Fisher et al., 2009). Therefore, a major challenge exists in reconciling values derived from public opinions on biodiversity with market-based values for commodities like timber (e.g. Aldred, 2002) (Fisher et al., 2009). Another example might be with integrating the valuation of tangible services like water provision with intangible aspects like spiritual benefits from a sacred forest (Fisher et al., 2009). Conservation costs can vary dramatically across different spatial extents (Ando et al., 1998; Ferraro, 2003; Balmford et al., 2003) and intangible ones like biodiversity's existence value are challenging to integrate (Naidoo et al., 2006).

Species' risk of extinction increases under a combination of threats such as environmental warming, overexploitation, and habitat fragmentation (Mora et al., 2007). Therefore, in conservation planning, economic costs are integral for conducting cost-benefit analysis (Margules & Pressey, 2000) and plans that take only a random sampling approach are unlikely to avert extinctions (Woodroffe & Ginsberg, 1998). Estimation methods for conservation costs traditionally have made use of aggregate measures such as total area, units, or constraints (e.g. Dobson et al., 1997; Araújo et al., 2002) that assume an equal cost (Naidoo et al., 2006). Initially conservation goals may be achieved at a relatively low cost and effort, but they exhibit diminishing returns as the cost and effort required to achieve complete conservation objectives escalates (Montgomery & Adams, 1994; Ando et al., 1998; Polasky et al., 2001; Naidoo & Adamowicz, 2005).

Migratory Response to Climate Change

Climate, particularly rainfall conditions, has an influence on mammal distribution and abundances in Africa (Owen-Smith, 1990; Mills et al., 1995). Tropical forests are more resilient to climate impacts than monoculture plantations or any artificial forest, especially when they are rich in biodiversity (Ravindranath, 2007). Species-climate envelope models (CEM) predict species distribution under differing climate scenarios but are highly sensitive to their chosen algorithm and context (Araújo et al., 2005; Hijmans & Graham, 2006). Also, the historical ranges of species hundreds or thousands of years ago is lacking in available data that may limit model predictions under future scenarios (Carvalho et al., 2021). Despite this, in the forthcoming decades CEMs predict the extinction of many African mammals (Thuiller et al., 2006) and sub-Saharan plant species (McCLean et al., 2005).

A key research challenge is in quantifying the impact of climate change on African tropical and subtropical plant and animal species that are important for current and future ES provision (Willis et al., 2013). Land transformation by humans restricts species range migrations under climate change and only then does the risk of extinction become a real possibility (Thuiller et al., 2006). Climate scenarios with low greenhouse gas emissions strongly limit the negative impacts of a changing climate (Allan et al., 2023). Also, during the AHP forest expansion was extensive but nonlinear as different species underwent range shifts at drastically different rates and directions (Watrin et al., 2009; Drake et al., 2011; Lézine et al., 2011). Therefore, understanding the nonlinear behaviour of African biota in response to climate change is essential in understanding the reliability of current and future predictions (Willis et al., 2013).

A major conservation challenge is to assess what effects climate and land use changes will have on species distributions under a range of environmental scenarios (Carvalho et al., 2021). Many terrestrial organisms are shifting their range to a higher latitude or elevation in response to climate change (Chen et al., 2011). Climate-driven crop migration and yield reductions have already been observed (Zhang et al., 2017; Sloat et al., 2020) and are projected for the future (King et al., 2018; Ceglar et al., 2019). As range shifts occur, new ecological relationships will be forged that may unpredictably transform the character of species interactions and fundamental ecosystem processes (Post et al., 1999; Walther et al., 2002; Schmitz et al., 2003) that could lead to species loss (Root et al., 2003).

Protected Areas and Forest Conservation

National parks and bio-reserves are key conservation tools that can be used to protect species and habitats within the confines of their boundaries (Thuiller et al., 2006). International agreements like REDD or the Kyoto Protocol are deemed necessary for managing congestible public goods (e.g. clean air) at a global scale, but differ from local-scale management strategies (e.g. Sandler, 1998) (Fisher et al., 2009). At a global scale cost per unit area for protection decreases with increasing area size, indicating that larger areas provide more benefits for a given investment (Naidoo et al., 2006). When a 'protected area' is formed and effectively managed, it can protect biodiversity and prevent forest degradation or conversion (Ravindranath, 2007). However, in designating a new national park as a protected area, decision makers are often faced with the challenge of balancing the equitable distribution of benefits with other objectives, such as education (Fisher et al., 2009).

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Climate change will alter the distribution of many habitats and component species, subsequently threatening the effectiveness of protected areas to meet conservation needs (Burns et al., 2003). Climate mitigation policies must recognise the influences of forests on climate, their differing spatial and temporal scales, and their long-term effectiveness and sustainability in a changing climate (Bonan, 2008). However, focusing ecological management and monitoring on provisioning or cultural services alone, threatens to overlook the various interactions between ES (Carpenter et al. 2006). Linking protected areas together is an adaptive strategy for flora and fauna to migrate in response to a changing climate (Ravindranath, 2007). Therefore, other objectives may include landscape management for preserving biodiversity and economic valuations such as ecosystem services and conservation value (Fisher et al., 2009).

Legally protected areas in Central Africa play a crucial role in forest preservation by largely withstanding the impacts of extractive industries, land conversion, and local hunting (Abernethy et al., 2016). However, evidence suggests that protected areas in equatorial Africa aren't able to withstand current poaching levels (Walsh et al., 2003; Bennett, 2011). Without adequate funding, governments may permit or overlook illegal and lucrative activities, such as logging, bushmeat hunting, and gold or diamond mining within protected areas (Wilkie & Carpenter, 1999). Logging concessions are an example of what might form a buffer zone in the vicinity of protected areas (Clark et al., 2009b). Without effective education, hunting regulations, buffer zones, and a reduction in logging leases, national parks risk becoming isolated pockets (Woodroffe & Ginsberg, 1998) of protected land.

Photo 8: Round-backed millipede on Fernan Vaz lagoon's banks.

Evaluating Costs

In developing countries there is often a lack of available data on land prices available at scales necessary for detailed conservation planning (Naidoo et al., 2006). Designating areas for conservation may elevate the value and, consequently, the price of adjacent lands, potentially increasing the likelihood of habitat conversion in nearby areas (Wu, 2000; Armsworth et al., 2006). For instance, protected areas may be considered as simply displacing logging to other areas (Barber et al., 2014). Which would mean increased pressure on non-protected areas and may even cause a net decrease in biodiversity (Wu, 2000; Naidoo et al., 2006). The challenge of lacking spatial data in valuations may be partly overcome by modelling cost-maps using inputs such as soil type, climate, current land cover, and proximity to roads (e.g. Chomitz et al., 2005) (Naidoo et al., 2006).

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Conservation initiatives will likely only work where the value of ES exceeds the opportunity costs forgone in abandoning the first best land-use option, plus the transaction costs (Wunder, 2008). The spatial variability of these costs may be partly accounted for by considering factors like the vulnerability or threats to habitats, or proximity to infrastructure and population density (Naidoo et al., 2006). The needs of many species may be approximated using just a few habitats (Krosby et al., 2015). Yet, protection, restoration, and conservation decisions are likely to consider habitats individually, despite some species being dependent on multiple priority habitats (Travers et al., 2021).

Habitat Fragmentation

Fragmented and isolated patches of forest are vulnerable to invasive species and hold little value for mammalian biodiversity (Gibson et al., 2013) and a reduction in the absolute range of a species will likely increase the risk of local extinction (Thomas et al., 2004; Thuiller et al., 2005). Possible landscape configurations may raise the problem of optimising where to restore over what to conserve (Hodgson et al., 2016). For example, why not high-quality 'refugia' habitats that have remained stable over long periods of time and are expected to be resilient towards climate change (Graham et al., 2010; Shoo et al., 2011). There are still many intact ecosystems about the globe that offer excellent conservation opportunities (Riggio et al., 2020) but the conservation of forests may benefit from directing efforts at large expanses of habitat (Morcatty et al., 2013).

About half of terrestrial protected areas are lacking in connectedness across the globe (Saura et al., 2018) with only 9.7% of Earth's terrestrial protected network actually considered structurally connected (Ward et al., 2020). Habitat fragmentation has been found to reduce biodiversity to anywhere between 13% and 75% while decreasing biomass and altering nutrient cycles (Haddad et al., 2015). Habitat loss and climate change pose a synergistic threat to biodiversity (Oliver & Morecroft, 2014; Synes et al., 2020; Mancini et al., 2022) and without proper designation, the likelihood of degradation or destruction increases and may severely impact the connectivity of habitat patches (Travers et al., 2021). Despite wildlife habitats across the world consisting mainly of small patches of poorly connected (semi-) natural habitat (Mancini et al., 2022), patches critical to species range expansion have generally not been prioritized and are underprotected (Travers et al., 2021).

Range Expansion (Primates)

Habitat loss and fragmentation, land use change and hunting, and the effects of commodity growth and trade is causing many primate species to approach extinction (Estrada et al., 2018). Climate change too will expose species, especially forest-dwelling primates to unsuitable conditions and limit their dispersal range (Carvalho et al., 2019). There are lots of uncertainties in modelling species responses to global environmental changes (Araújo & New, 2007; Thuiller et al., 2019). But over the past twenty years African apes have faced dramatic declines in suitable environment (Junker et al., 2012) in addition to large population losses (Walsh et al., 2003; Plumptre et al., 2016; Kühl et al., 2017; Strindberg et al., 2018) which is predicted to continue into the future due to climate change (Carvalho et al., 2021). Notwithstanding, different species have different dispersal abilities (Mancini et al., 2022) and range expansion may not necessarily be achieved through successful colonisation only (MacArthur & Wilson, 2001; Travers et al., 2021).

The limited dispersal capacity of many mammals means that range gains and losses can operate at timescales in the hundreds to thousands of years (Schloss et al., 2012). An ape generation lasts under 30 years (Plumptre et al., 2016; Kühl et al., 2017) which is unlikely to see any significant migration (Carvalho et al., 2021). Range expansion may be facilitated through improving the quality of existing habitat patches, increasing the permeability of the surrounding landscape, restoring degraded habitat, creating new 'stepping-stone' habitat patches, and by enlarging existing patches (MacArthur & Wilson, 2001; Crick et al., 2020; Synes et al., 2020). It is recommended that for great apes effective conservation strategies should be taxon specific and focus on existing protected areas, and proposing new ones based on habitat suitability models (Carvalho et al., 2021).

Technology

The drivers of rapid environmental change in Central Africa are primarily external economic forces that invest in resources like (Lewis et al., 2015) oil, minerals (Edwards et al., 2014a), timber (Hoare, 2015), agricultural land (Feintrenie, 2014), and wildlife products (UNEP & IUCN, 2013). Whereas, internally the surge in prices for ivory and other wildlife products since the mid-2000s has drawn well-funded criminal organisations to Africa, equipped with high-tech navigational tools, powerful rifles, and mobile communications (Abernethy et al., 2016). Synchronously, conservation technology has become increasingly important in the 21st century, with non-invasive techniques being at the forefront of these efforts (Piel et al., 2021). Therefore, conservation management must integrate a dynamic framework to strategically sequence conservation actions tailored to varying threat levels faced by different conservation targets (Naidoo et al., 2006).

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Technology enhances the field of conservation by providing innovative solutions for monitoring, early-warning damage mitigation, and policy implementation (Berger-Tal & Lahoz□Monfort, 2018). For instance, mapping forest carbon stocks may assist countries in assessing the likely carbon emissions avoided by implementing policies aimed at reducing deforestation and forest degradation (Saatchi et al., 2011a). Furthermore, modelled predictions of land suitability for crops is important to determine how climate-driven tree loss might influence governmental decisions and land use allocation (Abernethy et al., 2016). Therefore, better data collection is key to informing management decisions (Berger-Tal & Lahoz□Monfort, 2018), especially in a world where catastrophe mitigation can be financed by such mechanisms as crop-weather insurance (Campbell, 2009), grants, loans, and even bonds (Motlagh et al., 2024).

Obtaining high temporal resolution data series required for predictive climate models, may be facilitated by high-resolution paleoecological data and independent climate records (Willis et al., 2013). However, the sparse observational network of the Congo basin severely inhibits climate change projections (Delire et al., 2008; Washington et al., 2013) and tropical Africa is lacking in high-resolution paleoecological data such as pollen records (Ngomanda et al., 2007). Therefore it may be necessary to implement a network of radiosonde stations, weather radar (Washington et al., 2013), ground observations and drought manipulative experiments across the region (Zhou, 2014). Though even with strong predictive capabilities, unpredictable events such as a series of large volcanic eruptions, may cause several decades of substantial global and regional climate perturbations (Allan et al., 2023).

Wildlife

Automating the annotation of camera trap images/ videos (Rowcliffe et al., 2016; Chen et al., 2019) and sound recordings (Kalan et al., 2015; Spillmann et al., 2015, 2017) using machine learning, could allow researchers greater efficiency and accuracy when monitoring wildlife (Piel et al., 2021; Tuia et al., 2022). Machine learning algorithms can be incredibly powerful for analysing small datasets, but they can lose resolution when applied to larger and more complex datasets (Joppa, 2015). Camera traps and especially passive acoustic monitoring (PAM), may be set up to collect data remotely and are deployable in remote areas, making them ideal for studying primates (Enari et al., 2019; Crunchant et al., 2020) like gorillas. PAM may be used in dense tropical forests or at night, and can be deployed for months or years at multiple locations (Piel et al., 2021). But, while PAM offers more effective detection, high intracaller acoustic variability makes it difficult to implement automated detection (Zimmermann & Lerch, 1993; Bouchet et al., 2012).

Forests

Drones may be utilised to facilitate detailed land-cover classification and monitor changes in certain variables, but fall short of detecting individuals at distance or those obscured by overhanging canopy (Piel et al., 2021). More generally, remote sensors (RS) are mounted to drones, aircraft, satellites, environmental features, or animals and can capture detailed real-world data (Tuia et al., 2022). For example, project and national scale mapping of carbon stocks traditionally use satellite imagery and ground-based inventory samples of forest carbon density (Saatchi et al., 2007; Asner et al., 2010; Neigh et al., 2013; Hu et al., 2016). Light detection and ranging (LiDAR) and radar have also been used in RS to provide higher-resolution forest carbon density estimates (Drake et al., 2002; Asner et al., 2010; Saatchi et al., 2011b; Lang et al., 2021). RS are therefore essential in the systematic monitoring of forests to understand their response to climate change (Malhi & Wright, 2004; Asefi-Najafabady & Saatchi, 2013).

Local Context

Certain technologies may be viewed as "neoliberal conservation" whereby local communities are excluded from decision making (Arrighi et al., 2003; Igoe & Brockington 2007). However, there is a limitation of computational technology in conservation and a lack of funding and expertise in certain areas, such as with drones (Joppa, 2015). In the case of great apes in the wild, researchers have traditionally relied on non-invasive sampling, such as collecting faeces in order to carry out genetic analysis (Städele et al.w, 2022). Other conventional wildlife monitoring systems involve manual counting and observations of animals, which are labour-intensive, time-consuming, expensive (Witmer, 2005), and prone to errors or bias (Burghardt et al., 2012; McEvoy et al., 2016). For local communities to become responsible for the nature on their own lands, easy access to necessary skills and technology is required (Goldman, 2001) in addition to an appropriate strategy, such as crowdsourcing (Pimm et al., 2015). Thankfully, this has become possible with the advent of cheap prototyping and designing tools such as mobile phones, 3D printers, affordable PCBs, microcontrollers, and single-board computers (Berger-Tal & Lahoz□Monfort, 2018).

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Photo 9: Dwarf crocodile (Osteolaemus tetraspis) along the Mpivié river.

Summary

Central Africa is home to the Guineo-Congolian rainforests that have persisted in one form or another for millions of years. Gorillas have evolved alongside the forests but habitat loss and the bushmeat trade threaten their continued existence. Increasing temperatures from climate change is drying soil and excessive atmospheric CO_2 is altering vegetation compositions. Suitable agricultural land is adversely affected and so more forest is cleared in a feedback loop as fertile soils become increasingly rare. Logging practices could be made more sustainable but are generally destructive and pave the way for large-scale human habitation at the expense of biodiversity. In response to local, regional, and global pressures forest dependent and indigenous communities' are being driven to unsustainably exploit the forest too. Since their livelihoods and access to land are being drastically altered, developing a strategy to support these people must go beyond reducing pollution to meet climate goals. If correctly perceived and implemented then the conservation of tropical rainforest can contribute to both adaptation and mitigation actions.

Gabon and Cameroon have designated a number of national parks but wildlife and forest degradation is ongoing and climate change is making matters worse. Ecotourism offers a source of revenue for keeping ecosystems intact, but cannot compete with the economic returns of extractive industries like logging. Ecosystem Services are a school of thought that try to value the conservation of ecosystems but contrasting facts and opinions make defining ES difficult. Nevertheless, conservationists must consider a cost-benefit analysis when deciding where or what to attempt to conserve which means also understanding the effects of climate change and local contexts. The biota of protected areas may need to migrate in response to climate change and so linking protected areas together or conserving large expanses of habitat could help to avoid extinctions. Models are needed that can accurately predict land use change and species migration but high quality data is lacking for Central Africa. The emergence of cheap technologies that can be deployed to collect lots of data in the field is making conservation aims more achievable.

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Photo 10: Leaf-footed bug (Subfamily Coreinae) in logged primary rainforest along the Mpivié river.

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